

Load Profiling Method in Detecting non-Technical Loss Activities in a Power Utility

A. H. Nizar Z. Y. Dong M. Jalaluddin M. J. Raffles

Abstract — This paper presents comprehensive review on non-technical losses, load profiles and data mining techniques that currently being used in effort to minimize the non-technical loss activities. It also presents on the contributing factors in load profiles of electricity customers, using the knowledge discovery in databases (KDD) procedure, to determine the load profiles for different types of customers. In this paper, the customer load profiles are compared based on the type of day, by analysing their differences in their consumption behaviour. The objective of this study is to use the load profiling methods and data mining techniques to classify, detect and predict non-technical losses in the distribution sector, due to faulty metering and billing errors, as well as to gather knowledge on customer behaviour and preferences so as to gain a competitive advantage in the electricity market. This paper focuses mainly on the development of framework analysis of the customer behaviour.

Index Terms — Classification, Clustering, Customer Information System, Data Mining, Deregulation, Load Profiles, Non-technical Losses (NTL).

I. INTRODUCTION

RESTRUCTURING of electrical supply has begun. Electrical industry deregulation has been advancing in many countries, bringing with it enormous changes in the ways that distribution utilities handle information within their databases on customer information, metering and billing systems. The impact of deregulation will affect the way power utilities handle and manage their business. Utility companies, formerly vertically integrated under one umbrella, will become unbundled, with greater challenges facing them.

The paper is organized into sections as follows. A review on non-technical losses, load profiling and data mining techniques currently used are presented in the remainder of Section I. The methodological frameworks of load profiling including feature selection for time domain data and load shape data are elaborated in Section II. Section III describes a framework analysis of NTL and some preliminary results gathered using statistical outlier detection method. And finally, section IV ends with the conclusions.

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A. Nature of NTL

The power losses that affect electricity utilities are classified into two categories. These are firstly, technical losses, and secondly the non-technical losses (NTLs).

Technical losses occur during transmission and distribution and involve substation, transformer, and line related losses. These include resistive losses of the primary feeders, the distribution transformer losses (resistive losses in windings and the core losses), resistive losses in secondary network, resistive losses in service drops and losses in kWh meter [1].

NTLs, by contrast, relate mainly to power theft in one form or another. They are related to the customer management process and can include a number of means of consciously defrauding the utility concerned. Among these means are the following [2]:

- 1) Tampering with meters to ensure the meter recorded a lower consumption reading;
- 2) Arranging false readings by bribing meter readers;
- 3) Stealing by bypassing the meter or otherwise making illegal connections;
- 4) Arranging billing irregularities with the help of internal employees, including making out lower bills, adjusting the decimal point position on bills, or just ignoring unpaid bills.

In all such cases, electricity customers intentionally avoid paying their bills or are involved in pilferage, theft, and unauthorised use. The intention of the present study is to focus only on the detecting and predicting of NTL activities on the distribution level where deviations in customer behaviour can be detected.

By default, the electrical energy generated should equal the energy registered as consumed. However, in reality, the situation is different because losses occur as an integral result of energy transmission and distribution. Davidson [3] defined the energy losses in terms of the following equations.

Energy Losses

$$E_{Loss} = E_{Delivered} - E_{Sold} \quad (1)$$

Revenue Loss due to technical losses

$$C_{ComLoss} = U_{ElecCost} \times E_{Loss} + M_{MaintenanceCost} \quad (2)$$

Non-technical Loss

$$C_{NTL} = C_{ComLoss} - C_{TechnicalLoss} \quad (3)$$

Several studies have reported on calculating and breaking down the losses in power utilities, including [1], [4] and [5]. However, the concern of the present study is not with calculating or breaking down the losses into technical losses and NTLs. Rather, its focus is on detecting significant deviations in customers' consumption behaviour by means of the load shape diagram.

NTLs experienced by electricity utilities have an impact on a number of areas, including financial and economic outcomes and political stability. Financial impacts are the most critical for many utilities for they involve a reduction in profits, a shortage of funds for investment in improving the power system and its capacity, and the necessity for implementing means to cope with the losses [2]. More general economic impacts flow from power utilities that are experiencing increasing losses being associated with corruption and forms of internal political interventions.

The current methods of dealing with NTLs impose high operational costs and require extensive use of human resources. Several alternative efforts have been made in other countries to minimize such problems. Most concentrate on on-site technical inspections, which have the similar limitation of high operational costs, as well as taking up much human resource time. It is to overcome such limitations that this study proposes to investigate a method of monitoring and detecting deviations in customers' load profiles as an alternative to the current endeavors to mitigate NTLs.

The main problem in minimizing the NTL problem is actually resources availability [6] as well as operation cost involved. Therefore, this study presented a solution using data mining techniques in customer information system to identify, detect and predict NTL activities. There is a need to use customer information system database to extract hidden knowledge about customer behaviour and transform unstructured of personnel knowledge known by the investigation team into a well structured of loss investigation knowledge based.

B. Load Profiling for NTL analysis

A load profile is defined as an electricity load consumption pattern for a customer or a group of customers over a given period [7, 8]. Two examples of commercial and industrial customer load profiles which are illustrated in Figures 1 and 2 respectively have shown significant differences on their load shapes.

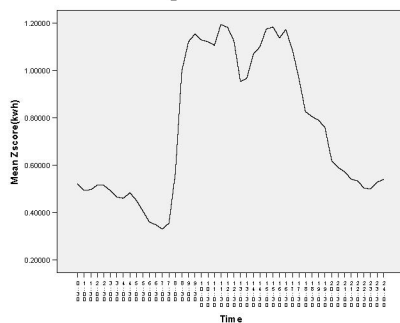


Figure 1: A Typical Commercial Load Profile

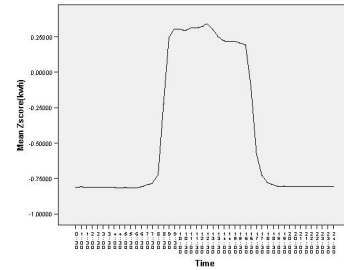


Figure 2: A Typical Industrial Load Profile

Load profile has been identified in many countries as an alternative solution that provides a satisfactory and cost effective approach compared to the interval metering solution, a solution that is known to be expensive and impractical for small low-voltage commercial and domestic customers. In addition to that, having a knowledge of customers' load profiles not only assists distribution companies in determining the demand price of electrical power, but it also provides better marketing strategies and improved efficiency [9].

C. Data mining techniques for NTL analysis

Due to the important needs to overcome the non-technical losses problem, several techniques and researches in detecting this kind of fraud and predicting the likelihood of this activities to happen has been increasing dramatically.

Based on the literatures gathered, we classify our reviews on fraud identification and detection techniques into two categories of business areas, which are:

The electricity business category: we review several data mining research studies on fraud identification and detection in electricity businesses, including rough sets [10], decision trees [11], Artificial Neural Network (ANN) [12], statistical-based outlier detection approach [13], wavelet-based feature extraction and multiple classifiers [14]. Most of these studies used data mining techniques directly to the customer database as the input.

The other types of businesses, we review several data mining techniques used other types of industries such as telecommunications [15], insurance [16], risk management [17] and credit card [18] businesses where data mining has been examined as a tool that enable the detection and prediction of fraud.

We also look into outlier detection research as it is one of the most important technologies in data mining and several successful application are always referred when talking about outliers such as credit card detection. Based on the literature gathered, we propose to detect the abnormalities in customer energy consumption through time series model. Variety techniques [19-22] have been applied to detect the outlier from that data features. The present research aims to examine means of detecting and predicting fraud in the electricity supply industry by means of outlier detection algorithm. In the following section, we present selected algorithms to be explored in our NTL framework analysis.

II. METHODOLOGY

A. A Composite Load Profiling Based NTL Analysis Framework

We propose a framework to analyse customer behaviour due to non-technical losses – A Composite Load Profiling based NTL (CLPNTL) analysis framework, as shown in the Figure 3.

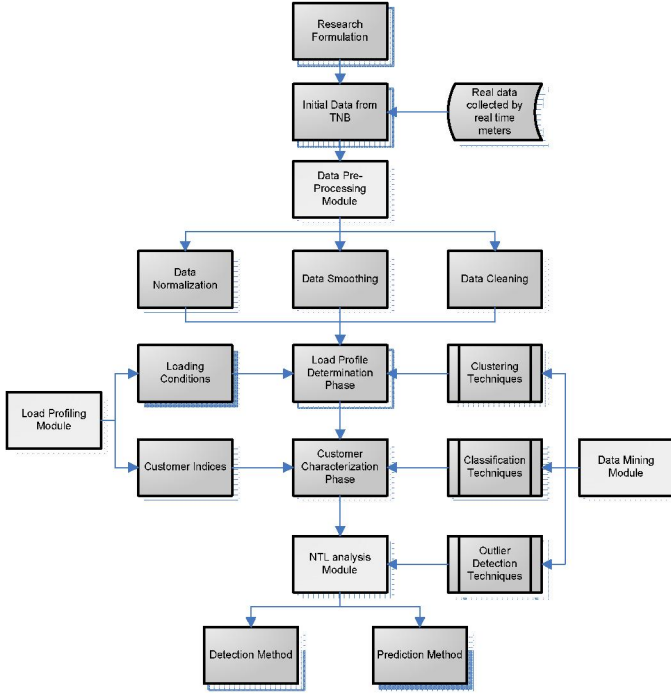


Figure 3: Composite Load Profiling based NTL (CLPNTL) Analysis Framework

The collected data is processed prior to load profiling module. The load-profiling module consists of two successive phases, which are load profile determination and customer characterization.

In load profile determination phase, a clustering technique is used to form K clusters in order to operate the partitioning of the initial L load patterns into the subsets $L^{(k)} \subset L$, for $k=1, \dots, K$ where each of them containing $n^{(k)}$ load patterns.

In customer characterization phase, once the customer classes have been formed, the representative load pattern $r^{(k)}$ of each customer class is derived from the load patterns in $L^{(k)}$. The load pattern obtained can be normalized to its peak value which resulting set $R = \{r^{(k)}, k=1, \dots, K\}$ contains the class representative load profiles.

Then the class representative load profiles where typical and untypical are used as a reference to monitor and investigate outliers that cause significant changes in customer behaviour in NTL analysis module using outlier detection techniques including statistical based, density based, distance based, model based and deviation based.

The final step is to compare and forecast the reference in the NTL analysis module using several techniques identified either by time series or by frequency domain. This module is also supported by data mining techniques and outlier detection techniques including Wavelet Transform (WT), Fast Fourier Transform (FFT) and Support Vector Machine (SVM) and time series.

B. Loading conditions

In the loading condition task, we separate our data into smaller data sets according to the loading conditions as identified in the table below:

Table 1: Loading conditions

Loading conditions	Items
Type of customer	Domestic
	Commercial
	Industrial
Location	Urban
	Rural
Voltage Level	Low Voltage
	High Voltage
Type of climate	Rainy
	Hot
Type of day	Week day
	Saturday (weekend)
	Sunday (weekend)
	Public Holiday

The need to set the loading conditions is crucial because we found out that it is obvious in the previous studies that different loading conditions yield different load shapes from every customer. Therefore, for the purpose of this study, we decided to develop a sample of representative load profiles for a commercial customer under the loading conditions category 'type of day' as set out in several figures below:

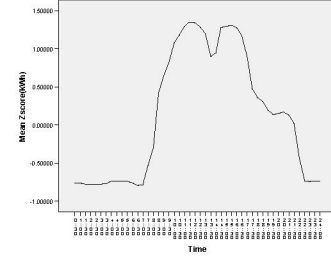


Figure 4: A typical load profile on a working day

This is typical load profile for m^{th} customer during working day which is based on average load consumption derived from Monday to Friday. It is found that in general this commercial customer starts his activity from 10 am and significant drop during lunch hour and peak again until 5 pm.

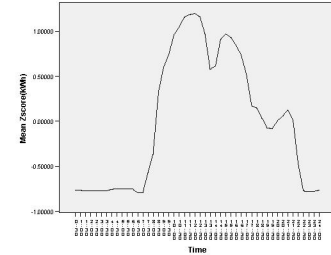


Figure 5: A typical load profile on Saturday

In some countries, alternate Saturday is a working day; therefore, it is found that the load curve for this commercial customer is almost identical to working day load curve with minor deviation.

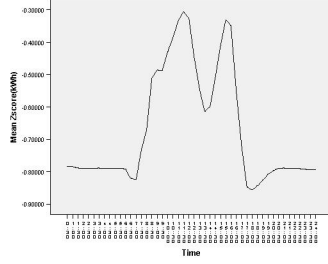


Figure 6: A typical load profile on Sunday

On Sunday, it is found that there is still electricity consumed for this commercial customer except that it is minimal.

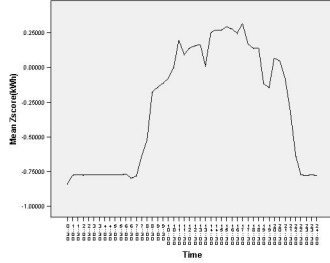


Figure 7: A typical load profile on a public holiday

For this particular commercial customer, it is found that this customer still operates their business during public holiday, however there are still slightly different curve happened during the public holiday.

From the various load diagram shown above, it is proven that a particular m^{th} customer can have various load patterns depending on the different loading conditions. Therefore, a careful analysis is required to be conducted. Besides the ‘type of day’ being identified as the loading conditions factor, we also consider weather conditions and climate changes in our study. We also look into type of customers, location of the customers and other factors including voltage level that may influence the customer load shape patterns.

Given a specific time domain data, we define the features of the m^{th} representative load patterns for $m=1,2,\dots,M$ as all of the normalised values obtained from the measurements. Hence, the set of features characterising the representative load pattern is defined as $C = \{c^{(m)}, m=1,\dots,M\}$, whose m^{th} component is represented by the vector $c^{(m)} = [c_1^{(m)}, \dots, c_H^{(m)}]^T$ where $H=48$ since we use 30 minutes interval in our measurement.

C. Feature Selection

We tested a set of sample time-domain and load shape data with the selected feature selection module and the results are listed as follows:

Table 2 : Average Accuracy of Different Search Strategies

Search Generation Strategy	Average number of features generated (time domain)	Average number of features generated (load Shape)	Average merit (time-domain)	Average merit (load shape)
Best-First	1.667	1.667	0.539	0.751
Genetic Search	2	2	0.539	0.751
Random Search	5	4	0.145	0.699
Greedy Stepwise	1.667	1.667	0.539	0.751

The average accuracies of different search strategies are listed in the table above. Obviously, Random Search has a better performance for time-domain data and load shape data. The summary for this result is illustrated in the figure below.

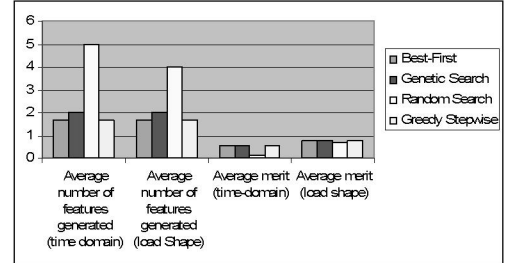


Figure 8: Average Merit and features numbers of search strategy.

Table 3: Average Accuracy of Different Evaluation Criteria

Evaluation Criterion	Average number of features (time-domain)	Average number of features (load shape)	Average merit (time-domain)	Average merit (load shape)
CFS	2.75	3.25	0.159	0.701
Consistency measure	3	2	1	0.552
Wrapper	0.333	0.333	0.453	0.999

The average accuracies of different evaluation criteria are also listed in the table above. Obviously, consistency measure is the best evaluation criterion compared to CFS and wrapper. However, the average merits of the consistency measure are the highest among all. This is also illustrated in the next figure.

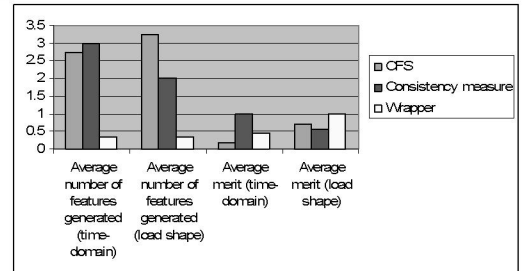


Figure 9 : Average merit and evaluation criteria

III. NTL FRAMEWORK ANALYSIS

We also propose a NTL detection framework as shown in Fig. 10. It illustrates the procedures in detecting NTL activities from the representative load profiles. This framework is a continuation from the composite load profiling based NTL (CLPNTL) analysis framework.

The first CLPNTL framework is able to generate the load profiles on the customers based on their similar characteristics, however in order to detect and predict NTL, further analysis is needed. We therefore propose a NTL detection framework using advanced data mining techniques = Data mining based NTL (DMNTLP) analysis framework as shown in Fig. 10. which illustrates the procedure in detecting NTL.

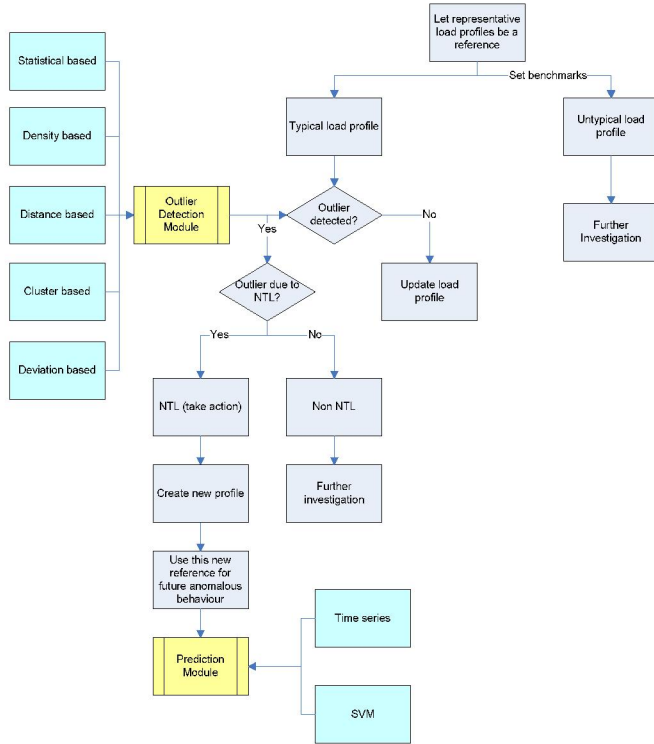


Figure 10: Data mining based NTL Detection Framework (DMNTL)

We summarize the main tasks in the DMNTL framework illustrated in Figure 10 as follows:

- We use the representative load profiles gathered from the load-profiling module as a reference. Generally, in the previous studies [23], most of the researchers rejected the untypical load profile due to insignificant behaviour. However, in our study, the untypical load profile is used as a benchmark for investigation with outlier detection techniques.
- We set up the benchmarks based on two types of load profiles, abnormal behaviour pattern and normal behaviour pattern. We use these two types of benchmarks as the training data. The abnormal patterns are to be investigated further.
- The typical load profile gathered is made as a reference in detecting NTL activities where it is compared with a new one. If there is no outlier, then the new load profile is

updated. But if there is deviation, then the new load profile is investigated using any outlier detection techniques including statistical based, density based, distance based, model based and deviation based.

- If the outlier is detected is confirmed due to NTL, then the new load profile is updated for an anomalous reference for forecasting purposes. Then this load profile is used as a reference for forecasting NTL activities for new load profile using forecasting techniques such as SVM and time series.
- We tested several samples using simple statistical-based outlier detection software and the results are follows:

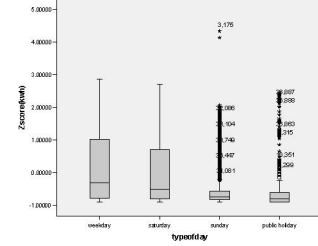


Figure 11 : Box plot based on type of day.

From the figure above, we investigated whether outlier exist based on type of day. We found that there are outliers on Sunday and Public Holiday.

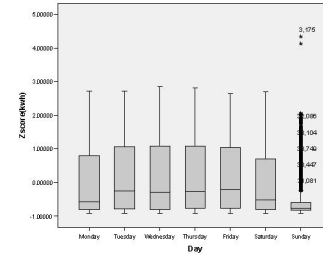


Figure 12 : Box plot based on days

From the figure above, we separated the data according to the days. We found that the outliers exist on Sunday.

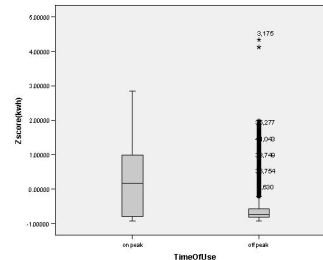


Figure 13 : Box plot based on time of use

Based on the above figure, we separated the data again according to the time of use, which is on peak and off peak. We found that the outliers exist during off peak hours during the weekday.

From all of the figures shown, we found that the outliers exist in these data especially during weekend and off peak hours. Therefore, our main investigation is to compare this method with other outlier detection methods.

IV. CONCLUSIONS

The major aim of this paper is to identify and examine the means for detecting and predicting NTL activities that utilize measured deviations in customer behaviour. Consequently, the determination of customers' load profiles and the allocation of individual customers to the classes identified as pertinent need to receive primary attention in the study.

Our main concern is to develop analysis framework of customer behaviour due to non-technical activities, based on these preliminary results, we investigate the outlier detected whether the outlier is mainly related to the non-technical activities. The on site investigation team can directly inspect the particular customer and confirm whether the outlier was caused by NTL.

If the detected outlier confirmed due to NTL activities, then we update this profile as a new reference for NTL pattern. We use the reference as a training input to built NTL forecasting model by considering other relevant external features as described in customer characterization section.

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VI. BIOGRAPHIES



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